

Dataset Description – IIT-H Campus Life Prediction

1 Overview

This dataset is designed to simulate student life at IIT Hyderabad and is used to predict the **mess duration**, which represents the time a student spends in the mess during a meal. The dataset captures multiple aspects of student behavior including daily schedules, environmental conditions, and academic pressure. These factors influence how long a student spends eating in the mess.

Initially, a dataset of 500 samples was generated using simulated data. The dataset was then expanded to 5000 samples by repeating existing samples and introducing Gaussian noise to selected features in order to simulate natural variability.

2 Dataset Attributes

The dataset contains the following features.

2.1 Student Information

Student ID

A unique identifier representing a student. It is generated to resemble IIT-style roll numbers that combine department, batch, and roll number.

2.2 Time and Context Features

Day of Week (0–6)

Represents the day of the week where:

- 0 = Monday
- 6 = Sunday

Meal Time

Indicates which meal the student is attending:

- 0 = Breakfast
- 1 = Lunch

- 2 = Dinner

Meal Type

Indicates whether the meal is vegetarian or non-vegetarian:

- 0 = Vegetarian
- 1 = Non-Vegetarian

Is Weekend

Binary variable indicating whether the day is a weekend:

- 0 = Weekday
- 1 = Weekend

Class Day

Indicates whether classes are scheduled on that day:

- 0 = Class day
- 1 = No class

Assignment Deadline

Indicates whether an assignment deadline exists on that day:

- 0 = No deadline
- 1 = Deadline present

Academic pressure is expected to influence eating behavior, as students may spend less time in the mess when they have assignments due.

3 Weather Features

Environmental conditions are included because weather can influence students' routines and their willingness to spend time outdoors or in common spaces.

Temperature (°C)

Generated using a normal distribution with an average of around 30°C to represent the typical climate of Hyderabad.

Humidity (%)

Simulated using a uniform distribution between 40% and 90%.

Wind Speed

Wind speed values are generated uniformly to represent varying atmospheric conditions.

Rainfall

Rainfall is simulated probabilistically. Most days have no rainfall, while some days contain moderate rainfall values.

Air Quality Index (AQI)

Air quality values represent pollution levels and are generated within a realistic range observed in urban environments.

Weather features help introduce environmental variability that may indirectly affect mess attendance and duration.

4 Lifestyle Features

Student lifestyle habits are also incorporated into the dataset.

Rising Time

Represents the time a student wakes up, measured as minutes from midnight. Values typically range between early morning and late morning.

Sleeping Time

Represents the time a student goes to sleep, also measured as minutes from midnight. This typically ranges from late evening to early morning hours.

These features reflect individual daily routines and may influence how students manage their meal schedules.

5 Target Variable

Mess Duration (minutes)

The target variable represents the amount of time a student spends in the mess for a particular meal.

The duration is generated using rules based on contextual factors:

- Breakfast duration is typically shorter because students may rush to classes.
- Lunch duration is moderate.
- Dinner duration is usually longer since students have fewer academic commitments.

Additional factors modify the duration:

- **Weekend effect**

Students tend to spend more time in the mess during weekends.

- **Assignment deadlines**

Students tend to eat more quickly when deadlines are present.

- **Heavy rain**

Rain may cause students to remain longer inside the mess.

- **Poor air quality**

Students may prefer to finish meals quickly under unfavorable environmental conditions.

These rules introduce logical relationships between the features and the target variable, which helps machine learning models learn meaningful patterns.

6 Dataset Expansion

The original dataset contained 500 samples. To obtain a larger dataset suitable for machine learning experiments, the dataset was expanded to 5000 samples.

The expansion process involved two steps:

6.1 Replication of Samples

Existing rows were repeated multiple times to increase the dataset size.

6.2 Gaussian Noise Injection

Random noise sampled from a Gaussian distribution was added to selected continuous features including:

- Temperature
- Humidity
- Wind speed
- Air quality
- Rising time
- Sleeping time
- Rainfall

Gaussian noise simulates small natural fluctuations in environmental and lifestyle variables while preserving the overall statistical properties of the dataset.

7 Final Dataset Characteristics

- Total samples: 5000
- Number of features: 14 input features
- Target variable: mess_duration
- Data type: Tabular dataset
- Features include behavioral, environmental, and academic factors

This dataset is suitable for experimenting with neural network models such as Perceptron, Adaline, and modern deep learning methods in order to study how different architectures learn patterns from real-world inspired tabular data.